MILITARY SPECIFICATION

SODIUM BIPHOSPHATE AND SODIUM CITRATE SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Sodium Biphosphate and Sodium Citrate Solution (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

FSC 6505

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(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be an aqueous solution containing in each 100 ml, 12.0 grams Sodium Biphosphate, U.S.P. (anhydrous basis), and 10.0 grams Sodium Citrate, U.S.P. (anhydrous basis) within the designated assay limits.

3.1.1 Sodium biphosphate content. Shall contain not less than 11.40 percent and not more than 12.60 percent w/v of Sodium Biphosphate (anhydrous basis) when tested as specified in 4.6.1.

3.1.2 Sodium citrate content. Shall contain not less than 9.50 percent and not more than 10.50 percent w/v of Sodium Citrate (anhydrous basis) when tested as specified in 4.6.2.

3.1.3 Identification. Solution shall comply with the identification tests for Citrate and Phosphate, as specified in 4.6.3.

3.1.4 Vehicle. The vehicle shall be Purified Water, U.S.P.

3.2 Ingredients. The Sodium Bisphosphate, Sodium Citrate and Purified Water used in the manufacture of the product shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.3 Volume in immediate container. The volume of solution contained in the immediate container shall be 133 ml $\pm$ 5 ml.

3.4 Delivery. Not more than 4 months shall have elapsed from the date of manufacture of the solution to the date of delivery to the Government.

3.5 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Package
Minor	II	2.5	Package

4.3.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

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TABLE II. For raw material and end item testing

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Sodium Biphosphate content	3.1.1	S-2	5
Sodium Citrate content	3.1.2	S-2	5
Identification	3.1.3	S-2	5
Volume in immediate container	3.3	S-2	5

Composite

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Solution not free from foreign matter.
102	Solution has crystal formation.
103	Solution not homogeneous.
104	Rectal tip not smooth and prelubricated.
Minor	
201	Exterior of immediate container is not free from contents and foreign matter.
202	Label not free from defacing marks and tears.
203	Tubing not of specified length.

1/ Inspection is not restricted to the classified possible defects listed above.

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4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay for sodium biphosphate. Add 50 ml of saturated NaCl solution to 25 ml (accurately measured) of the solution to be tested. Add 4 to 5 drops Phenolphthalein T.S. and titrate with 0.1 N. NaOH.

Calculations:

$$\frac{\text{Normality NaOH} \times \text{ml NaOH} \times 0.120 \times 100}{\text{ml of sample}} = \% \ NaH_2PO_4$$

4.7.2 Assay for sodium citrate. Shall be determined by total sodium assay using a flame photometer. The assay solution is prepared by diluting 5 ml of the solution for assay to 2,000 ml and further diluting a 10 ml aliquot portion to 100 ml. The sodium content shall be determined on the last dilution with a flame photometer.

Calculations:

The sodium content of the Sodium Biphosphate (as determined as specified in 4.7.1) is subtracted from the total sodium content and the difference is calculated to Sodium Citrate.

4.7.3 Identification tests.

4.7.3.1 For citrate. Evaporate one drop of the solution to be tested to dryness in a 50 ml beaker. Add 0.5 ml of acetic anhydride and 2 ml of pyridine. A yellow color indicates the presence of citrate.

4.7.3.2 For phosphate. Shall comply with the U.S.P. Identification Test for Phosphate ion.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable paragraphs of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One package containing one disposable enema unit, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate containers. The immediate container shall be a commercially available plastic bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. The immediate container shall have a permanently attached plastic tube approximately 18 inches in length. Shall be supplied with a smooth, pre-lubricated rectal tip.

5.2.2.2 Unit package. Each disposable enema unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Forty-eight unit packages shall be packed as specified in paragraph 5.2.9 of PPP-C-186.

5.2.3 Level C. Enema units shall be individually packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Two intermediate packages (96 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Two intermediate packages (96 units) shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The packaged enema units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.4 Marking.

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5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Labeling adhesion test shall not apply.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with paragraph 5.5.3 of PPP-C-186 except that the date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend:

"DO NOT PERMIT TO FREEZE".

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186 except that the date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend:

"DO NOT PERMIT TO FREEZE".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

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6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-890-1558	SODIUM BIPHOSPHATE AND SODIUM CITRATE SOLUTION, Disposable Enema Unit, 4-1/2 fl oz (133 ml)

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1510

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-37368 - SODIUM BIPHOSPHATE AND SODIUM CITRATE SOLUTION

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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